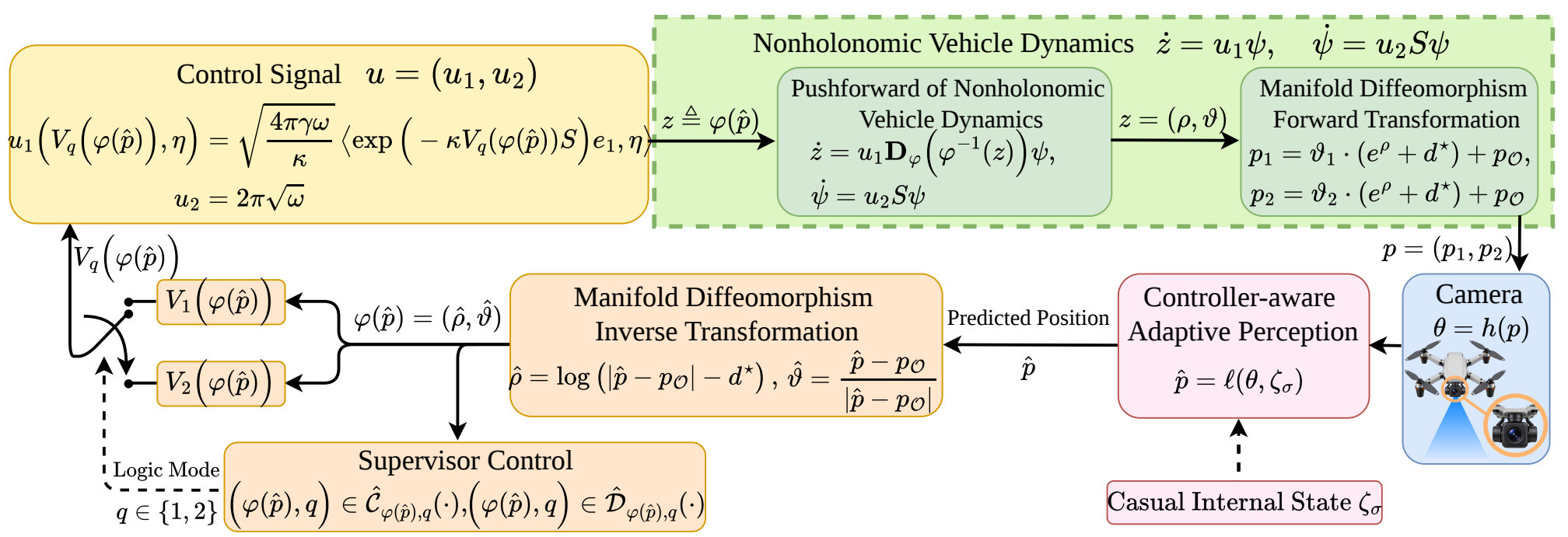

Control Signal $u = (u_1, u_2)$
$u_1\left(V_q\left(\varphi(\hat{p})\right), \eta\right) = \sqrt{\frac{4\pi\gamma\omega}{\kappa}} \left\langle \exp\left(-\kappa V_q(\varphi(\hat{p}))S\right) e_1, \eta \right\rangle$
$u_2 = 2\pi\sqrt{\omega}$
Nonholonomic Vehicle Dynamics $\dot{z} = u_1\psi, \quad \dot{\psi} = u_2 S\psi$
$z \triangleq \varphi(\hat{p})$
Pushforward of Nonholonomic Vehicle Dynamics
$\dot{z} = u_1 \mathbf{D}_\varphi\left(\varphi^{-1}(z)\right)\psi,$
$\dot{\psi} = u_2 S\psi$
$z = (\rho, \vartheta)$
Manifold Diffeomorphism Forward Transformation
$p_1 = \vartheta_1 \cdot (e^\rho + d^\star) + p_\mathcal{O},$
$p_2 = \vartheta_2 \cdot (e^\rho + d^\star) + p_\mathcal{O}$
$p = (p_1, p_2)$
Camera
$\theta = h(p)$
Controller-aware Adaptive Perception
$\hat{p} = \ell(\theta, \zeta_\sigma)$
Predicted Position
$\hat{p}$
Manifold Diffeomorphism Inverse Transformation
$\hat{\rho} = \log\left(|\hat{p} - p_\mathcal{O}| - d^\star\right), \ \hat{\vartheta} = \frac{\hat{p} - p_\mathcal{O}}{|\hat{p} - p_\mathcal{O}|}$
$\varphi(\hat{p}) = (\hat{\rho}, \hat{\vartheta})$
$V_q\left(\varphi(\hat{p})\right)$
$V_1\left(\varphi(\hat{p})\right)$
$V_2\left(\varphi(\hat{p})\right)$
Logic Mode
$q \in \{1, 2\}$
Supervisor Control
$\left(\varphi(\hat{p}), q\right) \in \hat{\mathcal{C}}_{\varphi(\hat{p}),q}(\cdot), \left(\varphi(\hat{p}), q\right) \in \hat{\mathcal{D}}_{\varphi(\hat{p}),q}(\cdot)$
Casual Internal State ζ_σ